

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO2: Foot printing Tools: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT3)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Report Analysis Questions

1. A security analyst performed a network scan on the target 192.168.1.10 using the Nmap tool. The report identified several open ports, including HTTP (80), SSH (22), and MySQL (3306), along with the operating system and service versions. Analyze the report and explain the security risks associated with the identified services. Discuss how an attacker could exploit these services and recommend suitable security measures to protect the target system. **(20 Marks)**

Commands :

1. nmap scanme.nmap.org
2. nmap -sV scanme.nmap.org
3. sudo nmap -O scanme.nmap.org
4. sudo nmap -A scanme.nmap.org

Observation :

- The host is alive.
- HTTP (80) is open.
- SSH (22) is open.
- SIP ports 5060 and 5061 are open.
- Operating system information identified.

Security Recommendations :

- Keep all software updated.
- Enable firewall protection.

- Use strong authentication mechanisms.
- Perform periodic vulnerability assessments.
- Conduct regular penetration testing.



Report : Question1_Nmap_Report.txt

```

kali@kali: ~
$ nmap --version
Nmap version 7.95 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.7 openssl-3.5.4 libssh2-1.11.1 libz-1.3.1 libpcres-10.46 libpcap-1.10.5 nmap-libndnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select

kali@kali: ~
$ nmap scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:40 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0023s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 987 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
2022/tcp  closed down
3389/tcp  closed ms-wbt-server
5002/tcp  closed rfe
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy

Nmap done: 1 IP address (1 host up) scanned in 88.31 seconds

kali@kali: ~
$ nmap -sV scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:43 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0084s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  tcpwrapped
53/tcp    closed domain
80/tcp    open  http        Apache httpd 2.4.7
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
8085/tcp  closed unknown

```

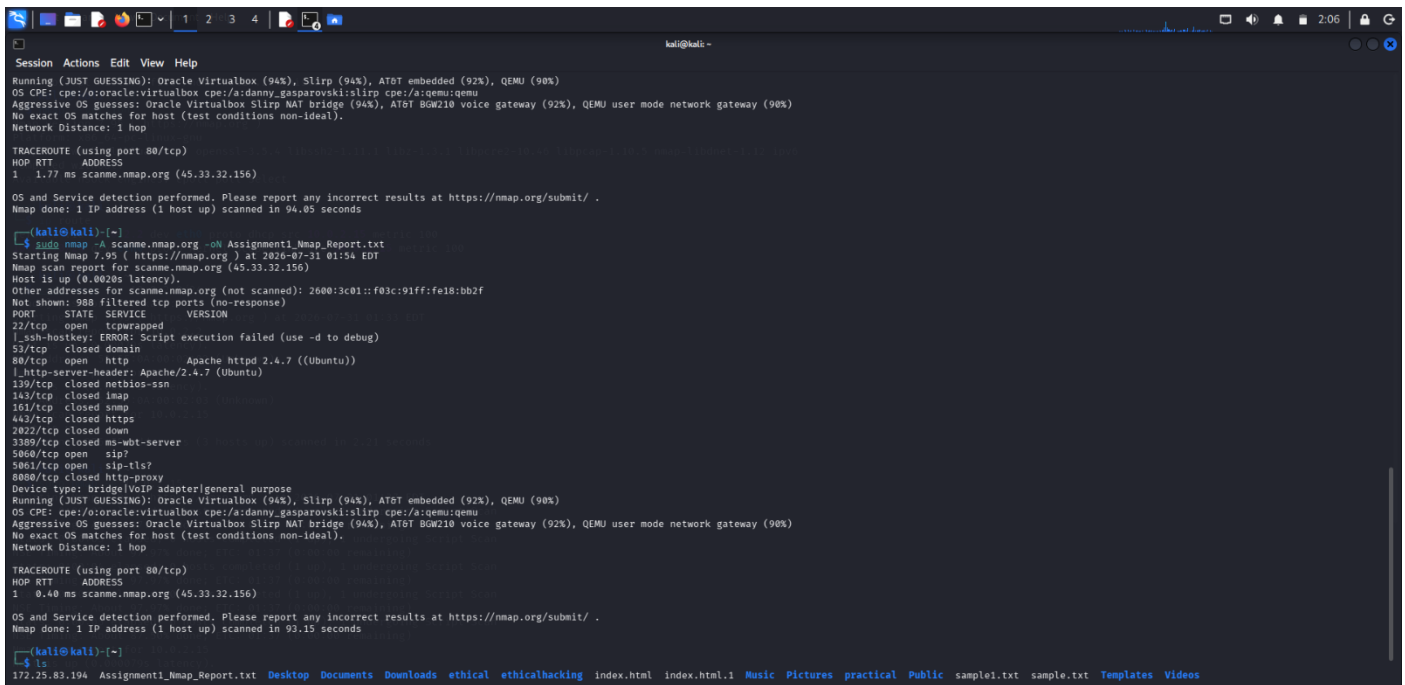
```

kali@kali: ~
$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:49 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.011s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 9.8p1 Ubuntu (Ubuntu)
53/tcp    closed domain
80/tcp    open  http        Apache httpd 2.4.7 ((Ubuntu))
837/tcp   closed mit-al-dev
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
580/tcp   closed isakmp
2022/tcp  closed down
3128/tcp  closed squid-http
3389/tcp  closed ms-wbt-server
5002/tcp  closed rfe
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
OS fingerprint not ideal because: Didn't receive UDP response. Please try again with -sSU
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 38.05 seconds

kali@kali: ~
$ sudo nmap -A scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:51 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0027s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 9.8p1 Ubuntu (Ubuntu)
53/tcp    closed domain
80/tcp    open  http        Apache httpd 2.4.7 ((Ubuntu))
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
Device type: bridge/VoIP adapter/general purpose
Running (JUST GUESSED): Oracle Virtualbox (94%), Slirp (94%), AT&T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny.gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (94%), AT&T BGM210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

```



```
kali@kali:~$ sudo nmap -A scanme.nmap.org -oN Assignment1_Nmap_Report.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-31 01:54 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0026s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  tcpwrapped
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
53/tcp    closed domain
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))
|_http-server-header: Apache/2.4.7 (Ubuntu)
139/tcp   closed netbios-ssn
143/tcp   closed imap
161/tcp   closed snmp
443/tcp   closed https
2022/tcp  closed down
3309/tcp  closed ms-wot-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
Device type: bridgeVoIP adapter[general purpose]
Running (JUST GUESSING): Oracle VirtualBox (94%), Slirp (94%), AT&T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle VirtualBox Slirp NAT bridge (94%), AT&T BGW210 voice gateway (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT      ADDRESS
1   1.77 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 94.05 seconds

kali@kali:~$ ls
172.25.83.194 Assignment1_Nmap_Report.txt Desktop Documents Downloads ethical ethicalhacking index.html index.html.1 Music Pictures practical Public sample1.txt sample.txt Templates Videos
```

2. A phishing awareness campaign was conducted for employees of ABC Technologies Pvt. Ltd. using the GoPhish tool. The report indicates that 40% of employees clicked the phishing link, 20% entered their login credentials, and several users downloaded a malicious attachment. Analyze the report and identify the security weaknesses demonstrated by the employees. Recommend suitable user awareness programs, email security controls, and organizational policies to reduce the risk of successful social engineering attacks.(20 Mark).

Procedure :

1. Created Email Template.
2. Created Landing Page.
3. Configured Mailtrap SMTP.
4. Created User Group.
5. Created Campaign.
6. Send phishing emails.
7. Viewed campaign statistics.

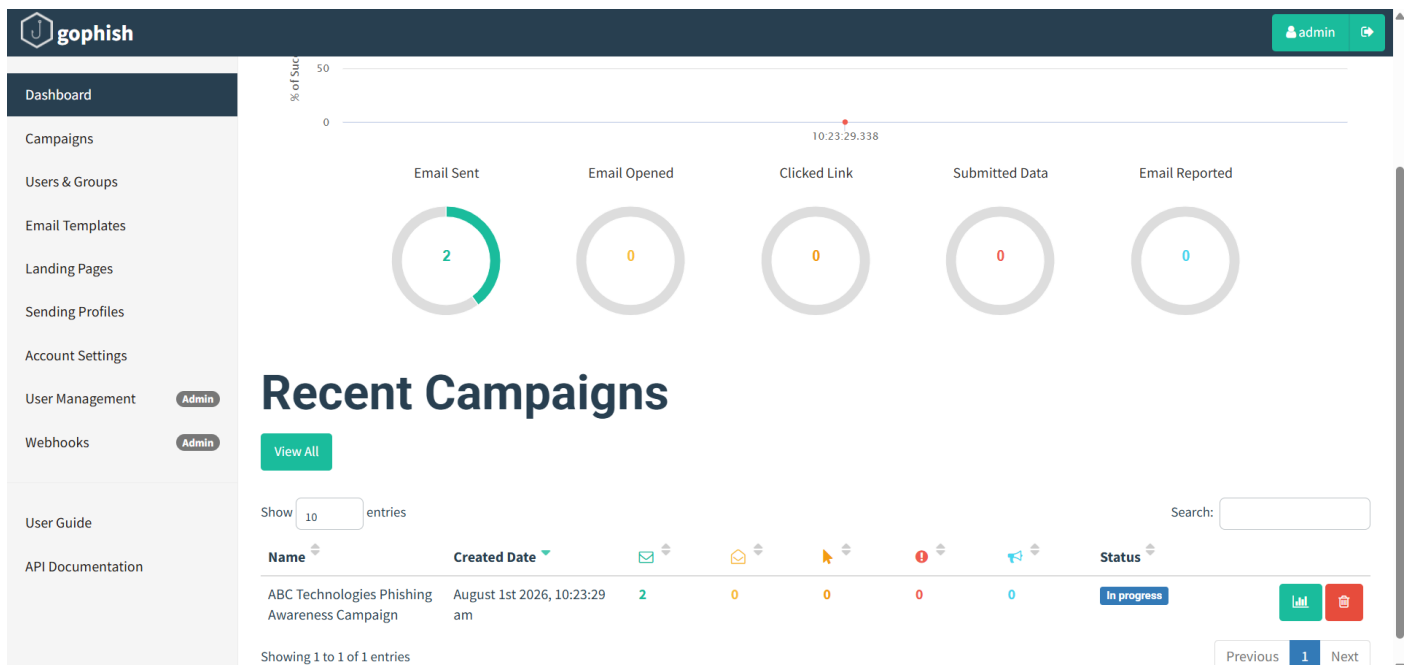
Email Security Controls :

- Enable Spam Filtering.
- Use Email Gateways.
- Configure SPF (Sender Policy Framework).
- Configure DKIM (DomainKeys Identified Mail).

- Configure DMARC (Domain-based Message Authentication, Reporting, and Conformance).
- Block dangerous attachments.
- Scan email attachments using antivirus software.
- Enable URL filtering.

Report :

ABC Technologies
Phishing Awareness C



3. A DNS assessment report generated using the Dig tool shows that the organization's DNS server permits a successful DNS Zone Transfer (AXFR), exposing internal hostnames, mail servers, and DNS records. Analyze the report and explain the security implications of the exposed information. Discuss how attackers could exploit these findings and recommend best practices to secure the DNS infrastructure..(10 Marks)

Command :

dig AXFR zonetransfer.me @nsztml.digi.ninja

Observation :

- DNS Zone Transfer succeeded.
- Internal DNS records were exposed.
- Mail server records were visible.
- Multiple hostnames and IP addresses were disclosed.

Possible Attacks :

An attacker can use the exposed information to:

- Perform reconnaissance.
- Scan exposed servers.
- Launch phishing attacks.
- Target mail servers.
- Exploit vulnerable web servers.
- Attempt brute-force attacks on exposed services.
- Prepare for advanced cyberattacks.

To Secure DNS Infrastructure :

- Disable DNS Zone Transfer for unauthorized users.
- Allow AXFR only between trusted DNS servers.
- Configure Access Control Lists (ACLs).
- Enable DNSSEC to protect DNS integrity.
- Regularly update DNS software.
- Monitor DNS logs.
- Remove unnecessary DNS records.
- Perform regular DNS security audits.

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
kali@kali:~$ dig -v  
Dig 9.20.15-2-Debian  
kali@kali:~$ dig AXFR zonetransfer.me @nsztml.digi.ninja  
;<>> Dig 9.20.15-2-Debian <>> AXFR zonetransfer.me @nsztml.digi.ninja  
;; global options: +cmd  
zonetransfer.me. 7200 IN SOA nsztml.digi.ninja. robin.digi.ninja. 2019100801 172800 900 1209600 3600  
zonetransfer.me. 7200 IN DNSKEY 256 3 7 AwEAAp0L+inQBYa2o13dIA24+eDFgVWwCOIMfCY3jLmgZABseuR8 ByhMQQxsoIOYU/7b3c8Tj2BwlQuqaxZ679QHSW7FK7D+bp/ASonB05 K5g3XEvphETJ9+8/X8Y971XaW9LmtJ6H4AAsrbgT2rg9K01PS1bOPM W8qULmQzK  
Tad9hVpC+uuzEDePMhoxSjPM+5QzVtfrf6y0z2yPcyYw 11A7P60Qv8vXsJdJy7QK4l0a8qzbybP0J07K3N1Dc8a0Z0XhS/R1L D0zxxZjY5V8FLwlmjeu07YAl11/AHQ7Dsu5FuZ52VxgCyR8nmX9GY0N 4S8vTF215eM+  
zonetransfer.me. 301 IN TXT "google-site-verification=ty2837JAUhA9f42sh9MgcC0Bt0X8moY10AV1MwXa"  
zonetransfer.me. 7200 IN MX 0 ASPMX.L.GOOGLE.COM.  
zonetransfer.me. 7200 IN MX 10 ALT1.ASPMX.L.GOOGLE.COM.  
zonetransfer.me. 7200 IN MX 10 ALT2.ASPMX.L.GOOGLE.COM.  
zonetransfer.me. 7200 IN MX 20 ASPMX2.GOOGLEMAIL.COM.  
zonetransfer.me. 7200 IN MX 20 ASPMX3.GOOGLEMAIL.COM.  
zonetransfer.me. 7200 IN MX 20 ASPMX4.GOOGLEMAIL.COM.  
zonetransfer.me. 7200 IN MX 20 ASPMX5.GOOGLEMAIL.COM.  
zonetransfer.me. 7200 IN A 5.196.105.14  
zonetransfer.me. 7200 IN NS nsztml.digi.ninja.  
zonetransfer.me. 7200 IN NS nsztml.digi.ninja.  
zonetransfer.me. 7200 IN CERT PKIX 0 0 MIDDvTCCaQUCHH5BGZor1YrXo5h90ipm0ADUE79MA0GCSqSs1B30QE CwUAMIGaMqswCQYVQSGEwJH0jEYMBYGA1UECAwPU291dGsgWw9ya3No aXJlMR1eAYDVQQHDA1aGwVnZm1lbGQvEjAqBgVBA6MURpZ2luaW5q YTEQM4A  
GAUUCwE9f3a21ZuZEVWYGA1UEAwEPU291dGsgWw9ya3NoaXJlMR1eAYDVQQHDA1aGwVnZm1lbGQvEjAqBgVBA6MURpZ2luaW5q YTEQM4A  
GAUUCwE9f3a21ZuZEVWYGA1UEAwEPU291dGsgWw9ya3NoaXJlMR1eAYDVQQHDA1aGwVnZm1lbGQvEjAqBgVBA6MURpZ2luaW5q YTEQM4A  
GAUUCwE9f3a21ZuZEVWYGA1UEAwEPU291dGsgWw9ya3NoaXJlMR1eAYDVQQHDA1aGwVnZm1lbGQvEjAqBgVBA6MURpZ2luaW5q YTEQM4A  
UEM0ImNoYUthPzF2FotVu0FrFvUWYEdQo Zs4ku0RIEx8pWHM2QUSM6KwVLLB/FE9566fovgxGwS33QaTKATAVCh D9KtLf6whh/ec+CGI0nbvGvjqZfmm/SymkdgEBW7q3+5ByfYrUoHc A2G030dkw6vUBt1j+3+145KzLXtVfFbClMPQvdf1gwp+pw+cW67o UBvFQZfZbaTp+9+v8F0B18f8fGj/  
Mae1n8r5Shm0ut8d3Pa0AQWtW3 H3Uv1nEboAMCAwEATANBgkqhkiG9w0BAQsFAAOCAQEAXopsfPv2/r7 tkKqFcsMmb7Z8d12U4nsBon+X7K5Nr6obrVAtm0+XwD8+2UgVYI08 URkL9LOX6VoYMMVrITIN8KR5sin5e3e4tZewsNGrVtkvbbKULV1CeBt DgmIm8rkZewU1uN0sq0t/wd3GUZe2CM9Dp  
KvPHPhC9uq3pVdASidVlp SApuij8K3L+VruzJvWveyTUKWASn115V7BgGf8039WMD1E1ZP81c AdmFy1fx+tuteM61z5xxk1tp0/eltb39cnKfQnrs81t0GZj3y8C3CCLY m4NUN200NAC0T7uzBeZ611FSNqQjvyFyomtPnaae0CYRHEw=  
acme-challenge.zonetransfer.me. 301 IN TXT "60a05HMDU9x5vvy7PaQvQwCUS5SgXvrbdirjePEsZi"  
sip.tcp.zonetransfer.me. 14000 IN SRV 0 0 5060 www.zonetransfer.me.  
14.105.196.5.IN-ADDR.ARPA.zonetransfer.me. 7200 IN PTR www.zonetransfer.me.  
asfdbauthns.zonetransfer.me. 7900 IN AFSDB 1 asfdbbox.zonetransfer.me.  
asfdbbox.zonetransfer.me. 7200 IN A 127.0.0.1  
asfdbvolum.zonetransfer.me. 7800 IN AFSDB 1 asfdbbox.zonetransfer.me.  
canberra-office.zonetransfer.me. 7200 IN A 202.14.81.230  
cndexec.zonetransfer.me. 300 IN TXT "; ls"  
contact.zonetransfer.me. 2592000 IN TXT "Remember to call or email Pippa on +44 123 4567890 or pippa@zonetransfer.me when making DNS changes"  
dc-office.zonetransfer.me. 7200 IN A 143.228.181.132  
deadbeef.zonetransfer.me. 7201 IN AAAA dead:beaf::  
dr.zonetransfer.me. 300 IN LOC 53 20 56.558 N 1 38 33.526 W 0.00m 1m 10000m 10m  
DCC.zonetransfer.me. 7200 IN TXT "AbcdEfg"  
email.zonetransfer.me. 2222 IN NAPTR 1 1 "p" "E2U+email" "" email.zonetransfer.me.zonetransfer.me.  
email.zonetransfer.me. 7200 IN A 74.125.206.26  
Hello.zonetransfer.me. 7200 IN TXT "Hi to Josh and all his class"  
home.zonetransfer.me. 7200 IN A 127.0.0.1  
info.zonetransfer.me. 7200 IN TXT "ZoneTransfer.me service provided by Robin Wood - robin@digi.ninja. See http://digi.ninja/projects/zonetransferme.php for more information."  
internal.zonetransfer.me. 300 IN NS intns1.zonetransfer.me.  
internal.zonetransfer.me. 300 IN NS intns2.zonetransfer.me.  
intns1.zonetransfer.me. 300 IN A 81.4.108.41
```

```
kali@kali: ~  
Session Actions Edit View Help  
Firefox ESR  
kali@kali:~$ dig AXFR zonetransfer.me @nsztml.digi.ninja > Assignment3_DNS_AXFR_Report.txt  
kali@kali:~$  
kali@kali:~$
```

Code of Conduct:

I K Sujith Sai Vighnesh (192311199) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Sujith Sai Vighnesh

RUBRICS FOR CALCULATION

Report Analysis

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Report Analysis and Interpretation	25%	Accurately interprets the security report, identifies all key findings, and explains their significance with excellent technical understanding.	Correctly interprets most findings with minor omissions or inaccuracies.	Identifies only some findings and provides limited interpretation.	Unable to interpret the report or identify key findings.
Identification of Security Risks and Vulnerabilities	25%	Correctly identifies all security risks, vulnerabilities, and possible attack vectors with clear technical justification.	Identifies most risks and vulnerabilities with minor errors.	Identifies only a few risks with limited technical explanation.	Fails to identify major risks or vulnerabilities.
Application of Cybersecurity Concepts and Countermeasures	30%	Applies appropriate ethical hacking concepts and recommends comprehensive, practical, and technically sound countermeasures.	Recommends suitable countermeasures with minor technical gaps.	Provides limited or partially appropriate recommendations.	Recommendations are inappropriate, incomplete, or missing.
Technical Communication and Documentation	20%	Response is well-organized, technically accurate, clearly presented, and uses appropriate cybersecurity terminology throughout.	Response is organized and understandable with minor documentation or presentation issues.	Response lacks clarity, organization, or contains several technical inaccuracies.	Response is poorly organized, incomplete, and difficult to understand.